Student Number: _____

The University of Melbourne

Semester 2 Assessment 2013

Department of Mathematics and Statistics

MAST30027 Modern Applied Statistics

Reading Time: 15 minutes

Writing Time: 3 hours

Open Book Status: Closed book

This paper has 8 **pages (including this page).**

Authorised Materials:

Hand-held electronic calculators

☑ The following items are authorised: (list here)

OR

☐ Students may have unrestricted access to all materials.

OR

☐ No materials are authorised.

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☑ **Yes** ☐ **No**

Instructions to Invigilators:

A 16-page script book need be supplied to each student

Instructions to Students:

1. There are 10 questions.
2. All questions may be attempted.
3. The number of marks for each question is indicated after the question.
4. The total number of marks available is 100.

Extra Materials required (please tick & supply)

☐ Graph Paper ☐ Multiple Choice Form ☐ Other (please specify)

1. Consider i.i.d. random samples $X_1, \cdots, X_n$ of X that has the *pdf*

$$f(x|\lambda, \theta) = \lambda(x + 1 - \theta)^{-\lambda-1} I(x \geq \theta),$$

where $I(\cdot)$ is an indicator function, and $\lambda > 1$ and $-\infty < \theta < +\infty$ are two unknown parameters to be estimated.

   (a) Specify the likelihood function $L(\lambda, \theta)$ given $X_1, \cdots, X_n$.
   (b) Find the MLE of θ based on the random samples $X_1, \cdots, X_n$.
   (c) Find the MLE of λ based on the random samples $X_1, \cdots, X_n$.
   (d) Find the MLE of $E(X)$ based on the random samples $X_1, \cdots, X_n$.

[2 + 3 + 3 + 4 = 12 marks]

2. Let $Y_1, \cdots, Y_n$ be independent random samples each having the *pdf*

$$f(y_i|\lambda_i) = 2\lambda_i y_i e^{-\lambda_i y_i^2} I(y_i > 0); \quad \lambda_i > 0; \quad i = 1, \cdots, n.$$

A GLM (generalised linear model) for $Y_1, \cdots, Y_n$ specifies $\lambda_i = \alpha + \beta x_i$, where x_i's are covariate observations, and (α, β) are unknown parameters to be estimated.

   (a) Find the log-likelihood function $\ell(\alpha, \beta) = \ln L(\alpha, \beta)$.
   (b) Find $\frac{\partial \ell}{\partial \alpha}$ and $\frac{\partial \ell}{\partial \beta}$ which specify the score function $U(\alpha, \beta)$.
   (c) Find $\frac{\partial^2 \ell}{\partial \alpha^2}$, $\frac{\partial^2 \ell}{\partial \beta^2}$ and $\frac{\partial^2 \ell}{\partial \alpha \partial \beta}$. Then list the Fisher information matrix $I(\alpha, \beta)$.
   (d) Given that $(\alpha^{(j)}, \beta^{(j)})$ is the current estimate in the process of finding the MLE of (α, β), we can update it to $(\alpha^{(j+1)}, \beta^{(j+1)})$ by the Newton-Raphson or Fisher scoring procedure. Specify the formula used in this update.

[2 + 2 + 5 + 3 = 12 marks]

3. Let $T_1, \cdots, T_n$ be i.i.d. random variables each of which has the *pdf*

$$f(t|\beta) = \frac{3}{\beta} t^2 e^{-t^3/\beta} I(t > 0); \quad \beta > 0.$$

Suppose we have observed $T_1 = y_1, \cdots, T_m = y_m$ and $T_{m+1} > c_{m+1}, \cdots, T_n > c_n$, where m is given, $m < n$, and $y_1, \cdots, y_m$ and $c_{m+1} \cdots, c_n$ are given numerical values. This implies that $T_1, \cdots, T_m$ are completely observed whereas $T_{m+1}, \cdots, T_n$ are partially observed in that they are right-censored. We want to use an EM algorithm to find the MLE of β.

   (a) Find the complete-data log-likelihood function $\ell(\beta) = \ln L(\beta)$.
   (b) In the E-step, we calculate

$$Q(\beta, \beta^{(k)}) = E\left[\ln L(\beta) \,\middle|\, T_1 = y_1, \cdots, T_m = y_m, T_{m+1} > c_{m+1}, \cdots, T_n > c_n;\ \beta^{(k)}\right]$$

where $\beta^{(k)}$ is the kth-step estimate of β. Show that

$$\begin{aligned} Q(\beta, \beta^{(k)}) &= \ln 3^n - n \ln \beta + \sum_{i=1}^{m} \ln y_i^2 + \sum_{i=m+1}^{n} E\left[\ln T_i^2 \mid T_i > c_i, \beta^{(k)}\right] \\ &\quad - \frac{1}{\beta} \sum_{i=1}^{m} y_i^3 - \frac{1}{\beta} \sum_{i=m+1}^{n} c_i^3 - \frac{(n-m)\beta^{(k)}}{\beta}. \end{aligned}$$

(*Note*: The result $E(T^3|T > c) = c^3 + \beta$ can be used without proof.)

(c) In the M-step, we maximise $Q(\beta, \beta^{(k)})$ with respect to β to find an update $\beta^{(k+1)}$ from $\beta^{(k)}$. Show that

$$\beta^{(k+1)} = \frac{1}{n}\left[\sum_{i=1}^{m} y_i^3 + \sum_{i=m+1}^{n} c_i^3 + (n-m)\beta^{(k)}\right].$$

(d) Suppose the sequence $\{\beta^{(k)};\ k = 1, 2, \cdots\}$ converges to the MLE $\hat{\beta}$ when $k \to \infty$. Show that

$$\hat{\beta} = \frac{1}{m}\left[\sum_{i=1}^{m} y_i^3 + \sum_{i=m+1}^{n} c_i^3\right].$$

[3 + 4 + 3 + 2 = 12 marks]

4. Let $X_1, X_2, \cdots, X_n$ be i.i.d. random variables from a distribution that has the *pdf*

$$f(x|\theta) = \theta(1-x)^{\theta-1} I(0 < x < 1); \quad \theta > 0.$$

Let $x_1, x_2, \cdots, x_n$ be observations of $X_1, X_2, \cdots, X_n$.

Now assume θ to be a random variable having a Gamma$(\beta.\alpha)$ distribution *a priori*. Namely the prior *pdf* of θ is

$$p(\theta|\alpha, \beta) = \frac{1}{\Gamma(\alpha)\beta^{\alpha}} \theta^{\alpha-1} e^{-\theta/\beta} I(\theta > 0); \quad \text{with} \quad E(\theta) = \alpha\beta \quad \text{and} \quad \text{Var}(\theta) = \alpha\beta^2.$$

(a) Find the posterior pdf of θ given $x_1, \cdots, x_n$.

(b) Find the Bayes estimator of θ when the squared loss function is used.

(c) Find the Bayes estimator of θ^2 when the squared loss function is used.

(d) Find a 95% Bayes credible interval for θ.

[5 + 2 + 2 + 1 = 10 marks]

5. A random number X is to be generated by the following acceptance-rejection (A-R) sampling algorithm:

1° Generate $U = u$ from Unif(0, 1) and $V = v$ from Unif(0, 1) independently.

2° If $u + v < 1$, deliver $X = -\ln(u)$; otherwise, go to 1°.

The *cdf* (cumulative distribution function) of X can be found using the following result:

$$P(X \le x) = P(-\ln(U) \le x \mid U + V < 1), \quad \text{for a given } x.$$

(a) Show that the *pdf* of X is $f(x) = 2e^{-x}(1 - e^{-x}), \quad x > 0.$

(b) Let N be the number of times steps 1° and 2° need be repeated before an X value is generated. What is the distribution of N and what is the mean $E(N)$?

[5 + 3 = 8 marks]

6. Let $X_1, \cdots, X_n$ be i.i.d. random variables from a distribution which has the *pdf*

$$f(x|\theta) = \theta x^{\theta-1} I(0 < x < 1); \quad \theta > 0.$$

It can be shown that the MLE of θ is $\hat{\theta} = n\left(-\sum_{i=1}^{n} \ln X_i\right)^{-1}$.

(a) Let $Y_i = -2\theta \ln X_i$, $i = 1, \cdots, n$. Show that the *pdf* of every Y_i is

$$g(y) = \frac{1}{2} e^{-y/2} I(y > 0).$$

Namely, $Y_1, \cdots, Y_n$ are i.i.d. $\chi^2(2)$ random variables. *Note* the *pdf* of $\chi^2(\nu)$ distribution is $g(y|\nu) = \left(2^{\nu/2}\Gamma(\nu/2)\right)^{-1} y^{(\nu/2)-1} e^{-y/2} I(y > 0)$, ν being the degrees of freedom parameter.

(b) Define $Q = 2n\theta\hat{\theta}^{-1}$. Use the result in (a) to show that Q is a pivotal quantity.
What is the distribution of Q?
Derive a 95% C.I. (confidence interval) formula for θ based on Q.

(c) Based on a sample of 50 observations from $f(x|\theta)$ given at the beginning of this question, 1999 nonparametric bootstrap samples are generated to estimate $\text{se}(\hat{\theta})$ and construct C.I. for θ. The following R output was obtained where x stores the generated $x_1, \cdots, x_{50}$ observations:

```
> library(boot)
> theta.bt=function(x,i){th.hat=(mean(-log(x[i])))^(-1)
        n=length(x)
        var.th.hat=(th.hat^4)*var(-log(x[i]))/n
        c(th.hat,var.th.hat)}
> bootQ6=boot(data=x, statistic=theta.bt, R=1999)
> bootQ6

ORDINARY NONPARAMETRIC BOOTSTRAP
Call:  boot(data = x, statistic = theta.bt, R = 1999)

Bootstrap Statistics :
    original     bias    std. error
t1* 5.916003  0.2084220   1.2431656
t2* 1.465046 -0.1327603   0.5451988

> boot.ci(bootQ6)

BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
Based on 1999 bootstrap replicates
CALL :   boot.ci(boot.out = bootQ6)

Intervals :
Level      Normal              Basic              Studentized
95%   ( 3.271,  8.144 )   ( 2.805,  7.792 )   ( 2.066,  8.578 )

Level     Percentile            BCa
95%   ( 4.040,  9.027 )   ( 3.638,  8.337 )
Calculations and Intervals on Original Scale

> qchisq(0.025,100); qchisq(0.975,100)

[1] 74.22193
[1] 129.5612
```

i. From the R output find the value of $\hat{\theta}$, the bootstrap estimate of $\text{se}(\hat{\theta})$, and the 95% BCa C.I. of θ. Briefly explain how the bootstrap estimate of $\text{se}(\hat{\theta})$ is calculated. What does BCa stand for?

ii. Calculate the 95% C.I. for θ using the formula obtained in (b) and the χ^2 percentile values in the R output.

[3 + 3 + 2 + 2 = 10 marks]

7. **Neyman-Pearson Lemma:** Suppose random variables $X_1, \cdots, X_n$ have the joint pdf $f(\mathbf{x}_n|\theta)$ with $\mathbf{x}_n = (x_1, \cdots, x_n)^T$. Let $\lambda(\mathbf{x}_n) = \dfrac{f(\mathbf{x}_n|\theta_1)}{f(\mathbf{x}_n|\theta_0)}$ be the likelihood ratio test statistic, and $C^* = \{\mathbf{x}_n : \lambda(\mathbf{x}_n) \geq k\}$ be a critical region for testing

$$H_0\colon\ \theta = \theta_0 \quad \text{vs.} \quad H_a\colon\ \theta = \theta_1,$$

where k is such that $P(\mathbf{X}_n \in C^* \mid \theta_0) = \alpha$. Then C^* is the most powerful critical region of size α for the test. Namely, for any critical region C such that $P(\mathbf{X}_n \in C|\theta_0) \leq \alpha$,

$$\pi_{C^*}(\theta_1) = P(\mathbf{X}_n \in C^* \mid \theta_1) \geq P(\mathbf{X}_n \in C \mid \theta_1) = \pi_C(\theta_1).$$

Question: Consider i.i.d. random samples $X_1, \cdots, X_n$ of X that has the *pdf*

$$f(x|\theta) = \frac{1}{24\theta^5} x^4 e^{-x/\theta} I(x > 0); \quad \theta > 0.$$

It is easy to see that $X_i \overset{d}{=} \text{Gamma}(\theta, 5)$, $i = 1, \cdots, n$; and $\hat{\theta} = \dfrac{1}{5n}\sum_{i=1}^{n} X_i$ is the MLE of θ.

(a) Determine an MP (most powerful) critical region for testing $H_0\colon\ \theta = \theta_0$ vs. $H_a\colon\ \theta = \theta_1$, where $\theta_1 > \theta_0 > 0$, at significance level α. Simplify the critical region as much as you can.

(b) Explain whether the critical region found in (a) is UMP (uniformly most powerful) of size α for testing $H_0\colon\ \theta \leq \theta_0$ vs. $H_a\colon\ \theta > \theta_0$.

(c) Derive a GLR (generalised likelihood ratio) test for testing $H_0\colon\ \theta = \theta_0$ vs. $H_a\colon\ \theta \neq \theta_0$. Simplify the derived critical region as much as you can.

[3 + 3 + 3 = 9 marks]

8. In a survey a random sample of 30 adults were asked the following two questions concerning the government's rights when it suspected that a terrorist act was about to happen:

(a) Do you agree the government has the right to tap people's telephone conversations?

(b) Do you agree the government has the right to stop and search people on the street at random?

The results of the survey are shown in the following table.

	Random Stop	
Tap Phone	Agree	Disagree
Agree	8	10
Disagree	2	10

An objective of the survey is to test $H_0\colon\ p_1 = p_2$ vs. $H_a\colon\ p_1 \neq p_2$, where p_1 is the probability of an adult agreeing the "Tap Phone" right and p_2 is that agreeing the "Random Stop" right.

(a) Let Y_{12} denote the top-right observation in the table. What is the distribution of Y_{12} when H_0 is true? Find the exact p-value of the test using McNemar's binomial test statistic.

(b) The following is the R output from the McNemar's χ^2 test. Explain how the test statistic value $\chi^2 = 4.0833$ is obtained.

```
> x=matrix(c(8,2,10,10),2,2)
> mcnemar.test(x, correct=T)

        McNemar's Chi-squared test with continuity correction
data:  x
McNemar's chi-squared = 4.0833, df = 1, p-value = 0.04331
```

[3 + 2 = 5 marks]

9. A study reported by A. F. Smith (1967), *Lancet*, **2**, p178, recorded the level of an enzyme, creatinine kinase (CK), for patients who were suspected of having a heart attack. The objective of the study was to assess whether measuring the amount of CK on admission to the hospital was a useful diagnostic indicator of whether patients admitted with a diagnosis of a heart attack had really had a heart attack.

   The enzyme CK was measured in 360 patients on admission to the hospital, with the measurements being discretized into 13 levels. After a period of time, a doctor reviewed the patients' records and determined which of the 360 patients had actually had a heart attack. The CK measurements (x), the corresponding numbers of patients admitted (n), and the corresponding numbers of patients who actually had a heart attack (y) are given in the following table.

x	20	60	100	140	180	220	260	300	340	380	420	460	500
n	90	39	38	35	21	20	19	14	19	15	7	8	35
y	2	13	30	30	21	19	18	13	19	15	7	8	35

   A logistic model $\ln \frac{\theta}{1-\theta} = \beta_0 + \beta_1 x$, where θ is the probability of a patient having a heart attack, is fitted to the data. The relevant results are provided in the following R output.

```
> ck.2 <- glm(y/n ~ x,family=binomial,weight = n,data=ck.dat)
> anova(ck.2, test="Chi")

Analysis of Deviance Table.  Model: binomial, link: logit. Response: y/n
Terms added sequentially (first to last)

     Df Deviance Resid. Df Resid. Dev  Pr(>Chi)
NULL                    12     311.29
x     1   283.15        11      28.14 < 2.2e-16

> 1-pchisq(28.14,11)

[1] 0.00308

> summary(ck.2)$coef

              Estimate  Std. Error   z value     Pr(>|z|)
(Intercept) -3.02835960 0.366977307 -8.252171 1.555422e-16
x            0.03510439 0.004081094  8.601712 7.853560e-18

> summary(ck.2)$cov.unscaled

             (Intercept)             x
(Intercept)  0.134672344 -1.293038e-03
x           -0.001293038  1.665533e-05
```

   (a) Use both Wald test and the (generalised) likelihood ratio test to assess the significance of the CK level x on affecting the probability of heart attack based on model `ck.2`. Report the test statistic value and the p-value for each test.

   (b) Does the model $\ln \frac{\theta}{1-\theta} = \beta_0 + \beta_1 x$ provide an adequate fit to the data? Why or why not?

   (c) Regardless of the result of (b), use the model $\ln \frac{\theta}{1-\theta} = \beta_0 + \beta_1 x$ to estimate the odds ratio of a heart attack when a patient's CK level increases by 50. Also find an approximate 95% C.I. for this odds ratio.

   (d) Use the model $\ln \frac{\theta}{1-\theta} = \beta_0 + \beta_1 x$ to estimate the probability of a heart attack when a patient has a CK level of 100. Also find an approximate 95% C.I. for this probability.

   [$2 + 2 + 3 + 4 = 11$ marks]

10. Three case-control studies were conducted to investigate the association between whether women had had any term pregnancy and whether they had ovarian cancer. The data are summarized in the following $2 \times 2 \times 3$ table.

Study	Term Pregnancies	Disease Status	
		Cancer	No Cancer
Study 1	None	31	93
	One or More	80	379
Study 2	None	39	74
	One or More	149	465
Study 3	None	35	95
	One or More	130	480

(a) A Mantel-Haenszel test is performed returning the following R output. Explain what this test is testing for and what the conclusion is at the significance level 0.05.

```
> q10=array(0,c(2,2,3))
> q10[,,1]=c(31,80,93,379)
> q10[,,2]=c(39,149,74,465)
> q10[,,3]=c(35, 130, 95, 480)

> mantelhaen.test(q10,correct=T)

        Mantel-Haenszel chi-squared test with continuity correction

data:  q10
Mantel-Haenszel X-squared = 9.8969, df = 1, p-value = 0.001656
alternative hypothesis: true common odds ratio is not equal to 1

95 percent confidence interval:
 1.176297 1.963098

sample estimates:
common odds ratio
         1.519601
```

(b) The data in the table are rearranged so that a log-linear model `mod1` can be fitted, with the results being given in the following R output. Give the mathematical formula (of the form $\ln(\lambda) = \cdots$) for model `mod1`.

```
> q10A.dat=data.frame(freq=c(31,80,93,379,39,149,74,465,35,130,95,480))
> q10A.dat$Study=factor(rep(c("Std1","Std2", "Std3"), each=4))
> q10A.dat$Pregnancy=factor(rep(c("None","OneOrMore"), times=6))
> q10A.dat$Disease=factor(rep(c("Cancer","NoCancer"),times=3, each=2))

> q10A.dat
   freq Study Pregnancy  Disease
1    31  Std1      None   Cancer
2    80  Std1 OneOrMore   Cancer
3    93  Std1      None NoCancer
4   379  Std1 OneOrMore NoCancer
5    39  Std2      None   Cancer
6   149  Std2 OneOrMore   Cancer
7    74  Std2      None NoCancer
8   465  Std2 OneOrMore NoCancer
9    35  Std3      None   Cancer
10  130  Std3 OneOrMore   Cancer
11   95  Std3      None NoCancer
12  480  Std3 OneOrMore NoCancer
```

```
> mod1=glm(freq~Study+Pregnancy+Disease+Study:Pregnancy+Study:Disease+Pregnancy:Disease,
 +  family=poisson, data=q10A.dat)
> anova(mod1, test="Chisq")

Analysis of Deviance Table

Model: poisson, link: log

Response: freq

Terms added sequentially (first to last)


                  Df Deviance Resid. Df Resid. Dev  Pr(>Chi)
NULL                                 11    1613.46
Study              2    22.82         9    1590.64 1.109e-05
Pregnancy          1   915.27         8     675.37 < 2.2e-16
Disease            1   649.15         7      26.22 < 2.2e-16
Study:Pregnancy    2     7.21         5      19.01  0.027225
Study:Disease      2     8.71         3      10.30  0.012829
Pregnancy:Disease  1     9.89         2       0.41  0.001658
---

> 1-pchisq(0.41,2)

[1] 0.8146473

> 1-pchisq(10.30,3)

[1] 0.01618084
```

(c) What assumption is made in order for the Mantel-Haenszel test in (a) to be valid? Test this assumption using the R output in (b) and draw the conclusion at the significance level 0.05.

(d) Test based on `mod1` the significance of the interaction effect `Pregnancy:Disease`, eliminating the effects of all other terms in the model. Comment on the implication of your result and how that is related to the result of (a).

(e) Test the adequacy of model

`Study + Pregnancy + Disease + Study:Pregnancy + Study:Disease`

at significance level 0.05, using the R output in (b). Comment on the implication of your result and how that is related to the results of (a) and (d).

$[2 + 2 + 3 + 2 + 2 = 11$ marks$]$

Total marks = 100

END OF EXAMINATION